TAPOMOY **ADHIKARI**

Residency Road, Richmond Circle, Bengaluru, Karnataka, India · +91 96098 00138
tapo3930@gmail.com · LinkedIn @tapomoyadhikari · Twitter @tapomoyadhikari

Tapomoy Adhikari is a Senior Research Fellow at Google and the Principal Researcher at Microsoft in the domain Artificial Intelligence and Machine Leaning. He is currently a school student of Griffins International School, Kharagpur. He lives in Kharagpur, West Bengal, India with his parents. His research career starts from June, 2018 at Google and now he has been given a designation of Senior Researcher at Google. His research areas mainly include, Artificial Intelligence, Machine Learning, Big Data Analysis, Cloud Computing, Web-based programming, Internet of Things, Neural Networks, Telecommunication, Networking, Linux Systems, and others.

He had published numerous research journals on Computer Science since 2018, which was proudly re-published by Google, Microsoft, IEEE, IIT Kharagpur, IIT Bombay, Cognizant Technology Solutions, and others. He is presently working on many new and innovative projects, out of which, he is going to publish some of his journals at IEEE.

He has been awarded with many certifications from the top technology giants. He, recently, had offered by Huawei, Microsoft, Google, Tata Consultancy Services, Cognizant Technology Solutions, Infosys India, Wipro India, Twitter and Yahoo. But he hadn't opted any other offers than Microsoft and Google. Microsoft and Google hired Tapomoy for performing the research in the field of Artificial Intelligence and Machine Learning. Google had offered money for working as an employee researcher at Google, but unfortunately, Tapomoy rejected all the paid offers and currently working at no-cost at Google, Microsoft, and Huawei consecutively.

As of 2023, he is recognized as one of the notable Junior-most Indian Computer Researcher in the field of Artificial Intelligence and Machine Learning. He is presently working on many projects and journals, and there is a great possibility from his side that, he may publish a powerful project for his nation, will submit a powerful thesis and may publish numerous books in the succeeding years.

EXPERIENCE

JULY 2018 – *PRESENT*

GOOGLE, SENIOR RESEARCHER

Conducting research studies utilizing methodologies pertaining to Artificial Intelligence, Machine Learning, Big Data Analysis, Cloud Computing, and other relevant disciplines.

OCTOBER 2022 – *PRESENT*

MICROSOFT CORPORATION, PRINCIPAL RESEARCHER

Conducting research using Machine Learning, Convolutional Neural Networks, Computer Vision, Networking Infrastructure, and other related areas.

FEBRUARY 2023 – *PRESENT*

AMAZON, SENIOR RESEARCHER

Appointed as the Senior Researcher with expertise in the fields of Artificial Intelligence, Big Data Handling, Machine Learning, Natural Language Processing, and other advanced programming techniques.

FEBRUARY 2023 – *PRESENT*

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY, JUNIOR RESEARCHER

Appointed as the Junior-most researcher of the institution under the domain of Artificial Intelligence and Machine Learning

JANUARY 2023 – *PRESENT*

HUAWEI TECHNOLOGIES, PRINCIPAL RESEARCHER

Appointed as the Principal Researcher for the domain Artificial Intelligence and Big Data Analysis.

FEBRUARY 2023 – *PRESENT*

BYTE TECHNOLOGIES, FOUNDER

Established a research laboratory named Byte Technologies (Byte-Tech IN) in India with the objective of creating an open-source research facility that would be accessible to all.

EDUCATION

SEPTEMBER 2020

GRIFFINS INTERNATIONAL SCHOOL, KHARAGPUR

Standard VII to *Present*

MARCH 2014 TO **SEPTEMBER 2020**

KENDRIYA VIDYALAYA AFS SALUA, KHARAGPUR

Standard I to VII

SKILLS

- **Programming**:
  C, C++, C#, Go, Python, Perl, Swift, Kotlin, PHP, Frontend (HTML, CSS, SCSS, Angular, React, React Native, others), Django, Conda, SQL, JavaScript, A#, A/ Sharp, MATLAB, R, Ruby, Jekyll, Malbolge
- Data Structure and Algorithm
- Statistics
- Calculus (Integral & Differential)
- Communication
- Learning
- Editor
- Author

PROJECTS

- Secured File Protocol for QoS Aware Systems and for organizational security
- QoS Aware Detector System using Augmented and Virtual Reality
- Realtime Chat Application
- Smart City using IoT for advanced agriculture
- Agricultural IoT: Enhanced farming in India
- Git Cloner

- Realtime Video Conferencing Environment
- Integrated C, C++, Perl & Swift Environment in OS X
- Sustainable Development Goal - Integrated using commercial IoT: Based on Linux Environment specializing in Fedora and RHEL 2.9 Linux
- Advanced Internet of Things: Campus Security System for Enhanced Security
- Kotlin, Java, Go, Objective C, Perl, Swift: Notes and To Do(s) Android & iOS Application, and Windows Software
- Swift: iOS-based IDE for advanced programming, named as AdvPro for iPhones
- Neural Machine Translation
- Generative Adversital Networks for Image Generation
- Python: Real-time face recognition using Deep Learning
- Exploration of the potential of reinforcement learning in Artificial Intelligence
- Implementation of a Secure Blockchain-based network
- Industrial IoT: Deep Learning for Predictive Maintenance in Industrial IoT
- Developing Autonomous Systems using Python based Artificial Intelligence
- Investigating the Use of Edge Computing for Improving IoT Applications
- Exploring the Use of Natural Language Processing for Sentiment Analysis in social media
- Investigating the Potential of Generative Adversarial Networks (GANs) for Computer-Generated Art and Design
- Development of a Smart Agriculture System using Drones and Computer Vision
- Evaluating the Effectiveness of Transfer Learning in Natural Language Processing
- Exploring the Potential of Augmented Reality for Enhancing Online Learning Experience
- Blockchain-based Secure File Sharing System
- Autonomous Vehicle Navigation using Machine Learning
- A System using Quantum Computing for Secure Tele-Communications
- Cloud-based Protected Document Collaboration System
- Implementation of Industrial IoT for the development in the Campus Security
- Setup of a Automated Emailer with voice instruction for Smart Offices
- Smart Home: A Sustainable Development Goal
- Stock Android 13 based Virtual Smartphone with integration of a physical smartphone
- Remotely accessible camera automation specially designed for home security
- Integrated Development Environment (IDE) supports all programming languages
- HO25 SSL high-security Socket Layer designed for corporation use
- Email server without any third-party client specially designed for Indian Army and its related field
- Telecommunication System using a Virtual Number dedicated for Indian Army
- PVR25 Socket Layer for additional security to the computer system
- Automatic Picture Capturing Application
- Dashboard for Smart Home Management

JOURNALS

- Development of Autonomous Systems using Python based Artificial Intelligence
- Investigating the Use of Edge Computing for Improving IoT Applications
- Exploring the Use of Natural Language Processing for Sentiment Analysis in social media
- Investigating the Potential of Generative Adversarial Networks (GANs) for Computer-Generated Art and Design
- Exploring the Potential of Augmented Reality for Enhancing Online Learning Experience
- Scalable Quantum Computer Architecture using Superconducting Qubits
- Development of a Cloud-based Infrastructure for Big Data Analytics [Published under Google and Microsoft, co-authored by IIT Kharagpur at IEEE Xplore]
- A Comparative Study of IP Address Representation Formats: Performance and Scalability Analysis [Published under Microsoft at IEEE Xplore]

- A Study of Computational Learning Theory and its Applications to Machine Learning [Re-published by Microsoft Corp.]
- Design and Implementation of Cloud-based Secured File Sharing System
- Investigating the Impact of Hyperparameter Optimization on Image Recognition with Deep Learning
- Dynamic Resource Allocation in Edge-Fog Computing for IoT: A Reinforcement Learning Approach
- Investigating the Feasibility of Edge Computing for Predictive Maintenance in Industry 4.0
- Real-Time Big Data Analytics for Transportation Optimization A Study of Predictive Modeling Approaches
- Investigating the Use of Graph Neural Networks for Text Classification
- Implementing Reinforcement Learning for Optimizing Resource Allocation in Cloud Computing

THESIS

- A Framework for Distributed Machine Learning: Overcoming the Challenges of Network Latency and Data Consistency
- Designing a Convolutional Neural Network for Image Recognition: A Comparative Study of Different Architectures and Training Techniques
- Designing a Low-Cost, Portable, and Non- Invasive Eye Tracking System for Clinical and Research Applications

BOOKS

- Exploring computational theory
- Impact of AI and ML in Computer Science
- Comparative Study of IP Address Representation Formats: Performance and Scalability Analysis
- Introduction to Quantum Computational Architecture
- Exploring the potential of Augmented Reality
- Introduction to Blockchain-based Networking System